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Meta-iPVP: a sequence-based meta-predictor for improving the prediction of phage virion proteins using effective feature representation

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Abstract
Phage virion protein (PVP) perforate the host cell membrane and eventually culminates in cell rupture thereby releasing replicated phages. The accurate identification of PVP is thus a crucial step towards improving our understanding of the biological function and mechanisms of PVPs. Therefore, it is desirable to develop a computational method that is capable of fast and accurate identification of PVPs. To address this, we propose a novel sequence-based meta-predictor employing probabilistic information (referred herein as the Meta-iPVP) for the accurate identification of PVPs. Particularly, efficient feature representation approach was used to generate discriminative probabilistic features from four machine learning (ML) algorithms making use of seven feature encodings. To the best of our knowledge, the Meta-iPVP is the first meta-based approach that has been developed for PVP prediction. Independent test results indicated that the Meta-iPVP could discern important characteristics between PVPs and non-PVPs as well as achieving the best accuracy and MCC of 0.817 and 0.642, respectively, which corresponds to 6–10% and 14–21% improvements over existing PVP predictors. As such, this demonstrates that the proposed Meta-iPVP is a more efficient, robust and promising for the identification of PVPs. The predictive model is deployed as a publicly accessible Meta-iPVP webserver freely available online at http://camt.pythonanywhere.com/Meta-iPVP.

Keywords Phage virion protein · Machine learning · Classification · Feature selection · Support vector machine · Meta-predictor

Introduction

Bacteriophages are viruses that can infect and thrive in bacteria. It can be found in several environments including soil, freshwater and marine. The infectious phage particle is essentially comprised of a nucleic acid component (i.e. either DNA or RNA) whereby they are encapsulated by a protein coat known as capsids [1]. Bacteriophages has an uncanny specificity towards a particular bacterial host species where they can irreversibly attach themselves to the surface of susceptible host and inject their genetic materials. They are able to persist in the host by one of two strategies namely via the lytic or lysogenic life cycle [2]. Bacteriophages represent a promising alternative to antiobiotics owing to the following properties: no toxicity toward human cells, causes no harm to normal flora and elicits action towards antibiotic-resistant bacteria [3].

Phage structural proteins (PVPs) are comprised of capsid proteins, tail proteins and phage particle enzymes. PVPs are

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responsible for orchestrating their interaction with bacterial hosts and as such represents a promising avenue to explore it as a potentially novel class of antimicrobial agents [4]. Current experimental approaches for the identification of PVPs from non-PVPs includes an arsenal of instrumentations consisting of mass spectrometry, sodium dodecyl sulfate polyacrylamide gel electrophoresis and protein array [5–7]. Despite the fact that these methods represent highly accurate and gold standard approach for PVP identification, they are not suitable in the analysis of PVPs at large scale owing to their laborious and costly nature. Thus, researchers have invested in efforts to develop computational models for predicting PVPs directly from their sequence information as a promising alternative.

Thusfar, several sequence-based models have been developed for PVP identification [8, 9] and this includes the seminal work of Seguritan et al. [10] as well as those from several other research groups including Feng et al. [11], PVPred [12], Zhang et al. [13], PVP-SVM [14], PhagePred [15], Tan et al. method [16], Ru et al. [17], Pred-BVP-Unb [18] and PVPred-SCM [19], as summarized in Table 1. Feng et al. [11] established the first benchmark dataset of experimentally confirmed 99 PVPs and 208 non-PVPs in their development of the PVP predictor using the Naïve Bayes (NB) as a function of amino acid composition (AAC) and dipeptide composition (DPC). In 2018, Manavalan et al. [14] established a widely used independent dataset consisting of 30 and 64 experimentally confirmed PVPs and non-PVPs, respectively, that was used to construct a sequence-based model named PVP-SVM, which uses support vector machine (SVM) in conjunction with 136 optimal features to yield an accuracy of 0.798 on the independent dataset. Shortly after, Arif et al. [18] employed an imbalance learning method (i.e. SMOTE) to overcome the inherent bias of the imbalanced benchmark dataset and proposed an predictive model built using SVM as a function of three different feature encodings known as Pred-BVP-Unb [11]. Particularly, this model provided the highest accuracy of 0.831 on the independent dataset. Until now, PVP-SVM and Pred-BVP-Unb represents the state-of-the-art PVP predictors with robust prediction accuracies. In spite of the reasonably high prediction accuracy of existing PVP predictors, it is still necessary to further improve the accuracy, generalization capability and transferability of the prediction model.

Motivated by the aforementioned considerations, this study proposes a novel sequence-based meta-predictor called the Meta-iPVP for differentiating PVPs from non-PVPs. The conceptual framework of the Meta-iPVP approach proposed herein for predicting PVPs is summarized in Fig. 1. Firstly, we established the largest dataset containing experimentally confirmed set of 313 PVPs and non-PVPs so as to improve the generalization capability and transferability of the model (Table 2). Secondly, this study makes use of a feature representation scheme encompassing the probabilistic information as derived from four machine learning (ML) algorithms as a function of seven feature encodings. Thirdly, the aforementioned probabilistic features served as input features to the SVM model. Finally, the model performance was augmented by making use of the genetic algorithm together with self-assessment-report (GA-SAR) for identifying the most informative feature set of size m. Upon objective assessment of an independent dataset, the Meta-iPVP (i.e. utilizing $m = 16$ selected probabilistic features) was found to display superior performance when compared to those of existing PVP predictors (e.g. PVPred [12] and PVP-SVM [14] and PVPred-SCM [19]) and widely used ML methods (e.g. NB, artificial neural network (ANN), random forest (RF)). Therefore, such comparison demonstrates the robust generalizability of the proposed model. To the best of our knowledge, Meta-iPVP represents the first meta-based approach for PVP prediction. We anticipate that Meta-iPVP may serve as a useful computational resource for high-throughput PVP prediction while also empowering experimental researchers in the discovery of novel PVPs. Finally,

Table 1 Summary of existing methods for predicting phage virion proteins

Method	Classifier	Sequence feature	Number dataset (TR/TS)	Web-server
Feng et al.'s method [11]	NB	AAC, DPC	307 (307/NA)	–
PVPred [12]	SVM	GDPC	307 (307/NA)	✓
PVP-SVM [14]	SVM	AAC, DPC, ATC, CTD, PCP	401 (307/94)	✓
PhagePred [15]	NB	GDPC	307 (307/NA)	✓[a]
Tan et al's method [16]	SVM	GDPC	401 (307/94)	–
Pred-BVP-Unb [18]	SVM	CT, SAAC, bi-PSSM	401 (307/94)	–
PVPred-SCM [19]	SCM	DPC	401 (307/94)	✓
Meta-iPVP	SVM	Probabilistic feature	626 (313/313)	✓

NB Naïve Bayes, *SCM* scoring card method, *SVM* support vector machine, *AAC* amino acid composition, *ATC* atomic composition, *bi-PSSM* bi-profile position specific scoring matrix, *CTD* chain-transition-distribution, *CT* composition and translation, *DPC* dipeptide composition, *GDPC* g-gap dipeptide composition, *PCP* physicochemical properties, *SAAC* split amino acid composition

[a]The webserver was not functional during our manuscript preparation

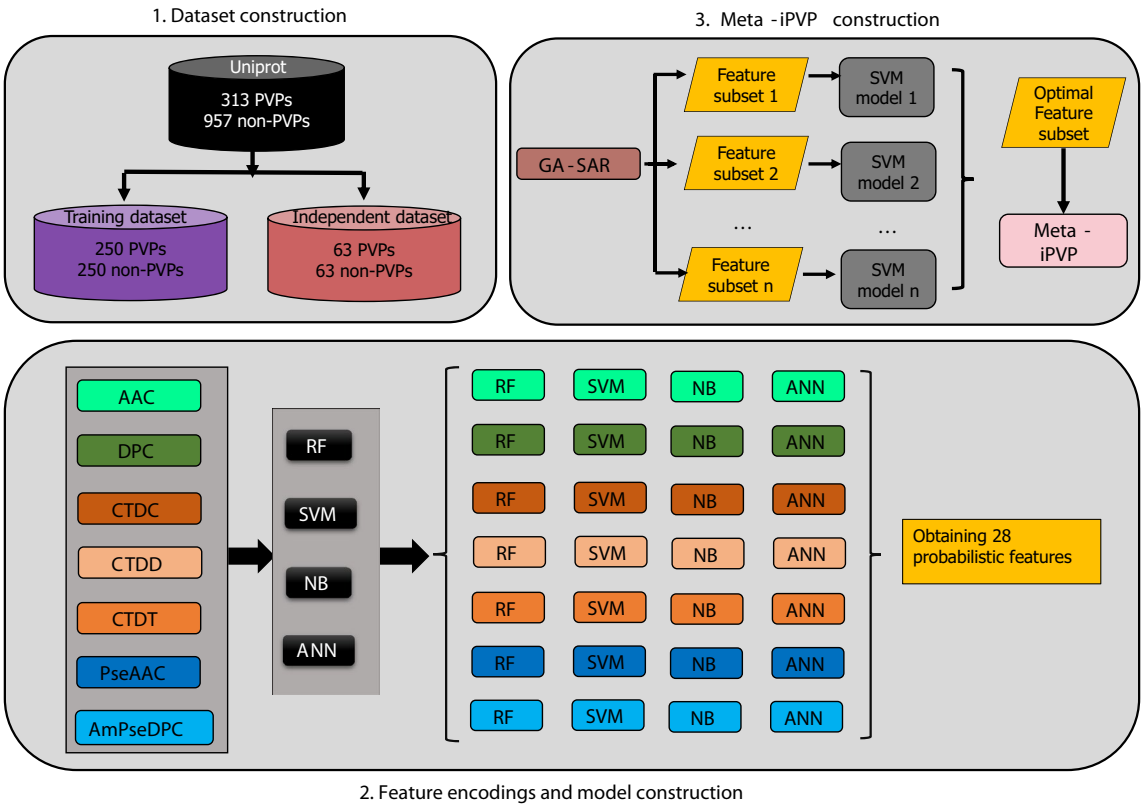


Fig. 1 Schematic framework of Meta-iPVP for predicting phage virion proteins

Table 2 Summary of existing and newly added benchmark datasets for predicting phage virion proteins

Dataset	Manavalan et al. [14]		Proposed dataset	
	PVP	Non-PVP	PVP	Non-PVP
Original	129	272	313	957
PVP-TR	99	208	250	250
PVP-TS	30	64	63	63

the model was deployed as the Meta-iPVP web server and made freely available online at http://camt.pythonanywhere.com/Meta-iPVP so as to lower the barrier to access of the predictive model.

Materials and methods

Benchmark datasets

Despite the fact that the benchmark dataset consisting of training (99 PVPs and 208 non-PVPs) and independent (30 PVPs and 64 non-PVPs) datasets [14] is an established and widely used dataset, however a larger number of samples is highly desirable in order to enhance the robustness and generalization capability of the model [20–24]. Thus, this study aims to grow the benchmark dataset by integrating established benchmark datasets [11, 14] with novel PVPs and non-PVPs obtained from the UniProt database (release 2019_11) [25]. Particularly, the set of novel PVPs and non-PVPs were collected by using the same keywords as obtained from the previous work [13]. Afterwards, the following steps were performed so as to obtain a high-quality dataset. Firstly, sequences containing ambiguous residues (e.g. X, B, U and Z) were excluded. Secondly, sequences containing <50 amino acids were excluded. Thirdly, sequences with $>40\%$ sequence similarity were removed using the CD-HIT program [26] using a threshold of 0.4 [27]. After performing such pre-processing procedures, a high-quality dataset consisting of 313 PVPs and 957 non-PVPs was obtained. In order to obtain a balanced dataset, the set of non-PVPs was reduced from 957 to 313 non-PVPs. Finally, a balanced benchmarking dataset constitutes equal numbers of PVPs and non-PVPs at 313 sequences each. In order to avoid overestimation in the prediction model, the benchmark dataset was randomly divided into training (named PVP-CV) and independent sets (named PVP-TS) using the 80/20 split ratio. In this work, the PVP-CV dataset consisted of 250 PVPs and non-PVPs, while the PVP-TS dataset consisted of 63 PVPs and non-PVPs (Table 2). In efforts to ensure the reproducibility of the proposed model, the PVP-TR and PVP-TS datasets used in

the construction of predictive models are available on GitHub at https://github.com/Shoombuatong2527/Benchmark-datasets/tree/master/PVP.

Feature representation

A given protein sequence (**P**) can be represented as:

$$\mathbf{P} = p_1 p_2 p_3 \cdots p_N \quad (1)$$

where p_i and N denotes the *i*th residue in protein **P** and the protein length, respectively. It should be noted that residue p_i is in the set of canonical amino acids (i.e. A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W and Y). Herein, sets of seven different types of sequence features were used to represent the protein sequences as feature vectors with fixed length and this consisted of AAC, DPC, CTDC, CTDD, CTDT, PseAAC and AmPseAAC. The definition of each sequence feature types can be briefly described hereafter

AAC and DPC represent the proportions of each amino acid and dipeptide in a given protein **P**. Thus, in terms of AAC and DPC features, the protein **P** can be expressed by vectors of 20D and 400D (dimension) spaces, respectively, as formulated by:

$$\mathbf{P} = [aa_1, aa_2, \ldots, aa_{20}]^{\mathbf{T}} \quad (2)$$

$$\mathbf{P} = [dp_1, dp_2, \ldots, dp_{400}]^{\mathbf{T}} \quad (3)$$

where T is the transposed operator while aa_1, aa_2,…, aa_{20} and dp_1, dp_2, …, dp_{400} are occurrence frequencies of the 20 and 400 native amino acids and dipeptides, respectively. It is well-known that AAC and DPC are one of the most interpretable and effective features for predicting and analyzing various types of proteins and peptides [19, 28–38].

As mentioned in previous studies [39–41] and shown in Equations (3)–(4), the sequence-order information of AAC and DPC may be completely lost. To address this issue, PseAAC and AmPseAAC were proposed by Chou [40]. According to Chou's PseAAC, the general form of PseAAC for a protein **P** is formulated by:

$$\mathrm{P} = [\Psi_1, \Psi_2, \ldots, \Psi_u, \ldots, \Psi_\Omega]^T \quad (4)$$

where the subscript Ω is an integer used for reflecting the feature's dimension. The value of Ω and the component of Ψ_u where u = 1,2, … , Ω are dependent on the protein or peptide sequences. The parameters of PseAAC (i.e., the discrete correlation factor λ and weight of the sequence information ϖ were estimated by using the optimization procedure. The dimension of the PseAAC feature is $20+\lambda \times \varpi$. The set of AmPseAAC features was introduced by Chou [40] for representing the hydrophobic and hydrophilic properties of proteins, which play important roles in their folding and interaction. The dimension of AmPseAAC feature is 20 + 2λ where the first 20 components accounts for the 20 basic AAC ($p_1, p_2, \ldots, p_{20}$) while the next 2λ denote the set of correlation factors that describes physicochemical properties encompassing the hydrophobicity and hydrophilicity along a protein sequence as formulated by:

$$\mathrm{P} = [p_1, p_2, \ldots, p_{20}, p_{20+\lambda}, p_{20+\lambda+1}, \cdots p_{20+2\lambda}]^{\mathrm{T}} \quad (5)$$

CTDC, CTDD and CTDT features are often used to describe the overall composition of amino acid properties of protein sequences. These features are derived from the CTD descriptor that is essentially described by a set of three descriptors consisting of Composition (C), Transition (T), and Distribution (D) [42]. The definition of the CTD descriptor can be briefly described as follows: (i) the composition descriptor calculates a percentage frequency of a particular amino acid property group in the protein sequence, (ii) the transformation descriptor describes the percentage frequency of the mutual transition of a polar group and a neutral group of a particular nature, and (iii) the distribution descriptor describes the distribution of the first, 25%, 50%, 75%, and 100% amino acids of a particular property within the protein sequence. Additional details of CTDC, CTDD and CTDT features can be found in the work of Xiao et al. [43]. In this study, five different sequence feature types were generated using the iFeature module in the Python environment [44].

Feature selection based on GA-SAR

Recently, the GA-SAR approach has been successfully implemented for predicting quorum-sensing peptides [45]. More specifically, GA-SAR utilizes a self-assessment-report (SAR) approach for constructing a profile that is used for reporting the usefulness of a feature pool based on the assumption that a good feature will be highly correlated with the output variable, but uncorrelated to each other [51,89]. Chromosomes in the GA-SAR is comprised of binary genes that indicates the occurrence of features in the feature set and parameters of model genes for tuning the classifier parameters. Herein, there are 531 binary genes and two 4-bit for encoding parameters C (2^{-8}, 2^{-7}, …, 2^{8}) and γ (2^{-8}, 2^{-7}, …, 2^{8}) of the SVM model. Interestingly, the GA-SAR algorithm coupled to the SVM model can automatically rank the most effective features and identify *m* informative features simultaneously.

Feature representation learning scheme

In the present study, we employed a newly developed feature learning scheme for extracting protein sequences as described by probabilistic information. This protocol has

been successfully implemented in the prediction of various proteins and peptides [28, 46–48]. To the best of our knowledge, this study represents the first to employ the feature learning scheme for the development of the PVP predictor. Procedure of the feature representation learning scheme is described in the paragraphs hereafter.

Firstly, the initial set of protein features was derived from seven feature encodings (i.e. consisting of AAC, DPC, CTDC, CTDD, CTDT, PseAAC and AmPseAAC which are expressed as fixed lengths of 20, 400, 39, 195, 39, 21 and 22, respectively). Parameters of PseAAC ($weight^1$ and $lamda^1$) and AmPseAAC ($weight^2$ and $lamda^2$) were optimized by varying the weight and lambda values from 0 to 1 and 1 to 10, respectively, using corresponding step sizes of 0.1 and 1 on the PVP-TR dataset and assessed by a tenfold CV procedure. In this work, values of $weight^1$, $weight^2$, $lamda^1$, and $lamda^2$ are 0.6, 0.8, 1 and 1, respectively. Further details on the parameter optimization can be found elsewhere [28, 38, 49, 50].

Secondly, the set of probabilistic features was generated from four ML algorithms (i.e. consisting of SVM, NB, ANN and RF) built using the aforementioned feature encodings. These four ML algorithms have previously been successfully implemented in various domains [14, 29, 30, 49–51]. In this step, 28 single-feature (SF) based models were established using the selected four ML algorithms and seven feature encodings so as to generate 28 probabilistic features. To obtain the most discriminative features, parameters of the 28 prediction models were optimized and implemented using the scikit-learn package (version 0.18) [47, 48]. Further details of SVM, NB, ANN and RF models and the protocol for parameter optimization are provided in the Supplementary Material.

Finally, in order to improve the feature representation ability, feature selection was applied for determining the optimal feature set. Herein, the GA-SAR algorithm was used to discern the minimal set of informative features while concurrently maximizing the prediction accuracy of the model [45]. In the experimental setup, GA-SAR was applied for the identification of *m* informative features from the aforementioned set of 28 probabilistic features, whereby the number of *m* is in the range of 10–20. It is hypothesized that if a feature is selected by GA-SAR, it is considered to be crucial for PVP prediction [45, 52, 53].

Performance evaluation

In order to evaluate the prediction ability of the model, four widely used metrics in binary classification were employed as follows:

$$\mathrm{Ac} = \frac{\mathrm{TP} + \mathrm{TN}}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})} \tag{6}$$

$$\mathrm{Sn} = \frac{\mathrm{TP}}{(\mathrm{TP} + \mathrm{FN})} \tag{7}$$

$$\mathrm{Sp} = \frac{\mathrm{TN}}{(\mathrm{TN} + \mathrm{FP})} \tag{8}$$

$$\mathrm{MCC} = \frac{\mathrm{TP} \times \mathrm{TN} - \mathrm{FP} \times \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}} \tag{9}$$

where Ac, Sn, Sp and MCC represents the accuracy, sensitivity, specificity and Matthews coefficient correlation, respectively. TP represents the number of correctly predicted proteins in the positive dataset (PVPs), TN denotes the number of correctly predicted proteins in the negative dataset (non-PVPs), FP represents the number of incorrectly predicted proteins in the positive dataset and FN represents the number of incorrectly predicted proteins in the negative dataset. Model comparison of the proposed model when those of previously described models was performed via the use of the receiver operating characteristic (ROC) curve of threshold-independent parameters. Correspondingly the area under the ROC curve (auAUC) was utilized to assess the prediction performance whereby AUC values in the range of 0.5 and 1 are indicative of random and perfect models, respectively.

Results

Cross-validation

In machine learning and pattern recognition, several studies [22, 54–60] and reviews [20, 23, 24, 61] have demonstrated that cross-validation is a crucial component for assessing the robustness and reliability of predictive models. Three approaches consisting of (i) sub-sampling test (2-, 5- or 10-fold cross-validation), (ii) jackknife test or also known as leave-one-out cross-validation (LOO-CV) and the (iii) independent (or external) testing dataset are often used to evaluate the prediction performance. In order to perform a fair comparison with those of previous related studies [11–17, 19], tenfold CV and independent tests were used.

Comparison of various classifiers on feature learning models

A series of comparative experiments was performed using various feature encoding methods (i.e. AAC, DPC, CTDC, CTDT, CTDD, PseAAC and AmPseAAC) for investigating

Table 3 Tenfold cross-validation results for the ten top-ranked single-feature based models having highest accuracies

Classifier	Ac	Sn	Sp	MCC	AUC
SVM-AAC	0.806	0.800	0.812	0.621	0.866
ANN-DPC	0.804	0.836	0.772	0.614	0.849
RF-AAC	0.794	0.772	0.816	0.596	0.874
ANN-AAC	0.792	0.804	0.780	0.590	0.845
ANN-PseAAC	0.786	0.748	0.824	0.581	0.847
ANN-AmPseAAC	0.780	0.736	0.824	0.567	0.844
RF-PseAAC	0.780	0.796	0.764	0.566	0.859
SVM-CTDT	0.780	0.764	0.796	0.566	0.835
RF-CTDT	0.770	0.748	0.792	0.548	0.826
RF-AmPseAAC	0.768	0.772	0.764	0.542	0.860

Table 4 Independent test results for the ten top-ranked single-feature based models having highest accuracies

Classifier	Ac	Sn	Sp	MCC	AUC
SVM-AAC	0.794	0.857	0.730	0.592	0.872
ANN-DPC	0.786	0.857	0.714	0.577	0.875
RF-AAC	0.722	0.730	0.714	0.445	0.815
ANN-AAC	0.762	0.841	0.683	0.531	0.882
ANN-PseAAC	0.762	0.825	0.698	0.528	0.861
ANN-AmPseAAC	0.738	0.794	0.683	0.479	0.818
RF-PseAAC	0.762	0.841	0.683	0.531	0.843
SVM-CTDT	0.754	0.762	0.746	0.508	0.848
RF-CTDT	0.746	0.730	0.762	0.492	0.820
RF-AmPseAAC	0.802	0.825	0.778	0.604	0.840

their contributions using different ML algorithms. Each feature encoding method was employed in the development of PVP predictors using four widely used ML classifiers (i.e. RF, NB, ANN and SVM). In total, 28 SF-based models were obtained for PVP prediction and their performances were evaluated via 10-fold CV and independent tests and are listed in Supplementary Tables S1 and S2, respectively. Moreover, Tables 3 and 4 lists the ten top-ranked SF-based models affording high accuracies as evaluated via tenfold CV and independent tests, respectively. Table 3 shows that the five top-ranked SF-based models consisted of SVM-AAC, ANN-DPC, RF-ACC, ANN-AAC and ANN-PseAAC with corresponding accuracies of 0.806, 0.804, 0.794, 0.792 and 0.786, respectively. It can be seen that two models from the five top-ranked SF-based models had MCC values exceeding 0.6 namely SVM-AAC and ANN-DPC had values of 0.621 and 0.614, respectively. In the meantime, three models from the five top-ranked SF-based models had MCC values exceeding 0.8 and this included SVM-AAC, ANN-DPC and ANN-AAC with values of 0.800, 0.836 and 0.804, respectively. As for the independent test results, Table 4 shows that the SVM-AAC model still performs better than most SF-based models and yields an accuracy of 0.794 with MCC and AUC of 0.592 and 0.872, respectively. From Table 4 it can be seen that the accuracy of the RF-AmPseAAC model was slightly higher than that of SVM-AAC. However, considering from both tenfold CV and independent tests, it could be stated that the SVM model built using the AAC feature (called the baseline model) is superior to those of other SF-based models.

Meta-iPVP construction using feature representation learning scheme

In regards to the feature representation learning scheme, probabilistic information from the predicted PVP probabilistic features were generated from the 28 SF-based models. In this section, we discuss which feature sets contributed the most to PVP prediction. Comparative analysis of predictive abilities of SVM models built with 28D and *m*D probabilistic feature vectors was conducted. As mentioned above, GA-SAR was applied for identifying *m* informative probabilistic features from the original 28 probabilistic feature vectors. Due to the non-deterministic characteristics of the GA-SAR algorithm [45], ten individual experiments were performed so as to generate ten different feature subsets. Supplementary Table S3 shows that the use of 28 probabilistic features provided an accuracy of 0.814 with MCC and AUC of 0.637 and 0.873, respectively, as assessed by 10-fold CV test. Furthermore, Supplementary Table S4 shows that the highest accuracy of 0.846 with MCC and AUC of 0.698 and 0.874, respectively, was obtained from experiment #6 in which the optimal feature set had $m = 16$ informative probabilistic features. These results showed that the accuracy and MCC values obtained using the optimal feature set were approximately 3% and 4% higher than using the original 28 probabilistic features, indicating that the optimal feature was essential for differentiating PVPs from non-PVPs. Further details on the optimal feature set is provided in Supplementary Table S5. For conciseness, we will refer to the proposed meta-predictor based on the SVM model and built with the optimal feature set as Meta-iPVP.

Analysis of new feature representation

In this section, we used the GA-SAR algorithm to rank and identify informative features on the 28 probabilistic features. The GA-SAR algorithm offers an easy way to rank and identify informative features using frequency [45]. Here, 30 individual experiments were performed to generate 30 different feature subsets. It could be stated that the feature having 30 and 0 feature usages represents the best and worst descriptors, respectively, in terms of the importance. As can be noticed in Fig. 2, the three most informative features were derived from SVM-CTDT, RF-CTDD and ANN-DPC with

corresponding usage values of 30, 29 and 24, respectively. Accuracies of SVM-CTDT, RF-CTDD and ANN-DPC were 0.780, 0.694 and 0.804, respectively, as assessed by tenfold CV. Remarkably, the accuracy obtained using Meta-iPVP was approximately 7–14% higher than these three SF-based models thereby indicating that the proposed model is superior to the 28 SF-based models.

As can be seen in Table 5, improved performances of the proposed Meta-iPVP was achieved owing to the robustness of the proposed feature representation learning scheme. To investigate this further, principal component analysis (PCA) [29, 30, 49, 51] was used to visually compare the distributions (i.e. scores plot) of PVPs (red) and non-PVPs (blue) as derived from various features including AAC (A), DPC (B), CTDC (C), CTDD (D), CTDT (E), PseAAC (F), AmPseAAC (G), 28 probabilistic features (H) and optimal feature set (I) as shown in Fig. 5. Herein, PCA analysis was performed in the Python environment [44]. It should be noted that, more overlap between red and blue circles indicates that the feature is less capable distinguishing PVPs from non-PVPs. Figure 3h and i reveal that the new feature representation can pick out informative features. These results also revealed that why the proposed Meta-iPVP is superior to the 28 SF-based models.

Comparison of Meta-iPVP with related ML predictors

The proposed Meta-iPVP was compared with those of other meta-predictors. For fair comparison, models were developed using the same benchmark dataset (PVP-TR and PVP-TS datasets) and optimal feature sets as mentioned above. Table 5; Fig. 4 summarize the performance comparison of the Meta-iPVP with those of existing meta-predictors as developed using the optimal feature set and evaluated by tenfold CV and independent tests. Interestingly, the accuracy and MCC values obtained from the Meta-iPVP were approximately 3–5% and 4–10% higher than those of other meta-predictors as assessed by tenfold CV. For independent test results, we observed a similar tendency as that of tenfold CV results as also shown in Table 5. The Meta-iPVP exhibited a better performance than other meta-predictors and the 28 SF-based models, achieving 1–2% and 2–5% improvements in accuracy and MCC, respectively, thereby justifying that the proposed Meta-iPVP represents the state-of-the-art approach for PVP identification.

Comparison of Meta-iPVP with the state-of-the-art predictors

To evaluate the predictive efficiency and robustness of the proposed Meta-iPVP, we further tested and compared its performance by using the PVP-TS dataset that is comprised of 63 PVPs and non-PVPs as summarized in Table 2. As described in the Introduction section, several computational models have been developed for identifying PVPs. In this study, we compared the proposed Meta-iPVP with three existing sequence-based PVP predictors namely the PVPred [12], PVP-SVM [14] and PVPred-SCM [19], because these are three well-known and newly developed PVP predictors. It should be noted that results for PVPred, PVP-SVM and PVPred-SCM were obtained by using protein sequences from the PVP-TS dataset as predicted via the web servers of the three aforementioned

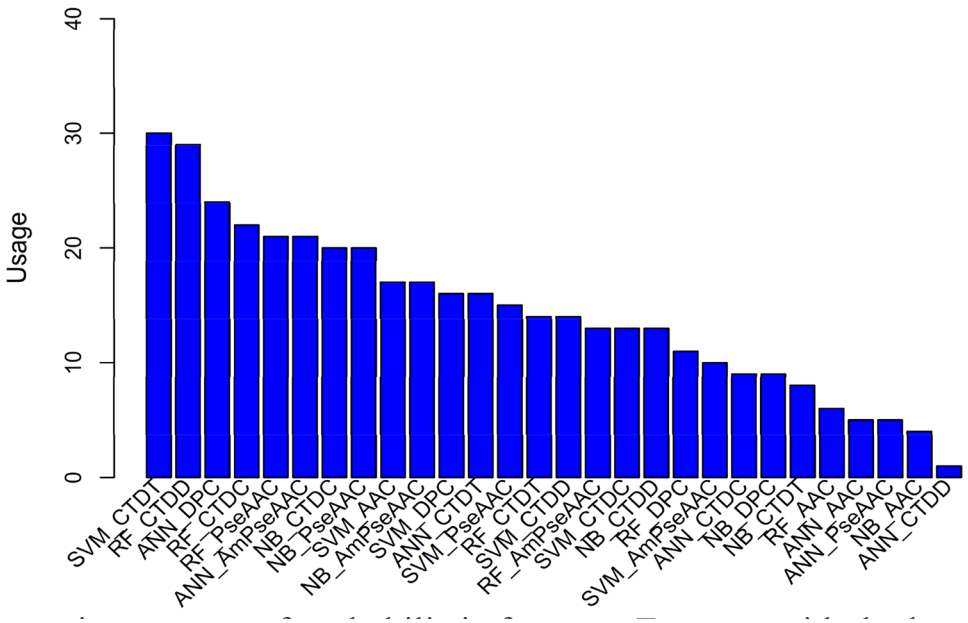


Fig. 2 Feature importance of probabilistic features. Features with the largest value of feature usage was considered to be the most important

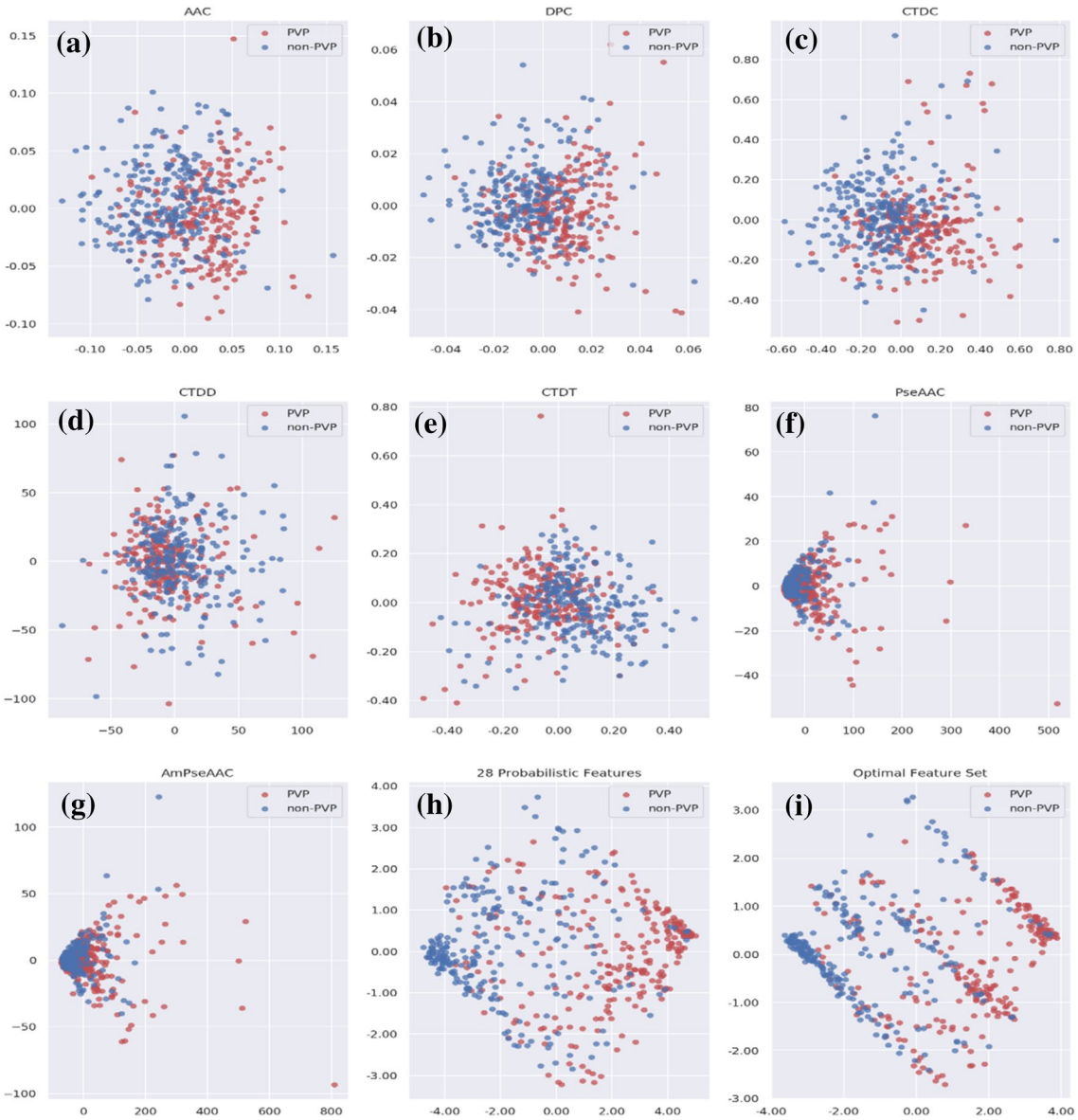


Fig. 3 PCA scores plot showing the distribution of PVPs (red) and non-PVPs (blue) as derivedfrom various features consisting of AAC (**a**), DPC (**b**), CTDC (**c**), CTDD (**d**), CTDT (**e**), PseAAC(**f**), AmPseAAC (**g**), 28 probabilistic features (**h**) and optimal feature set (**i**)

predictors. Table 6 lists the performance comparisons between Meta-iPVP with those of existing PVP predictors as evaluated by tenfold CV and independent tests. Table 6; Fig. 5 shows that the Meta-iPVP produces robust performance as deduced by accuracy, sensitivity, specificity, MCC and AUC of 0.817, 0.889, 0.889, 0.642, and 0.870, respectively. Remarkably, the accuracy, sensitivity and MCC of Meta-iPVP demonstrated markedly improved performance with 6–10%, 18–22% and 14–21%, respectively, higher values than those of existing PVP predictors. These results demonstrated that Meta-iPVP could outperform existing methods. This can be ascribed to the following reasons: (i) The proposed model was developed using a large dataset consisting of 313 PVPs and non-PVPs. To the best of our knowledge, this represents the largest dataset used in the development of a PVP predictor. (ii) A wide array sequence-based features (i.e. including residue composition, physicochemical properties, pseudo-amino acid composition and distribution patterns of amino acid properties) were used in model construction. (iii) The proposed Meta-iPVP model represents the first model to be developed using probabilistic information as derived from the proposed feature learning scheme (i.e. thereby reducing the high-dimensional feature space onto a low-dimensional feature space) that demonstrated robust predictive ability for identifying PVPs.

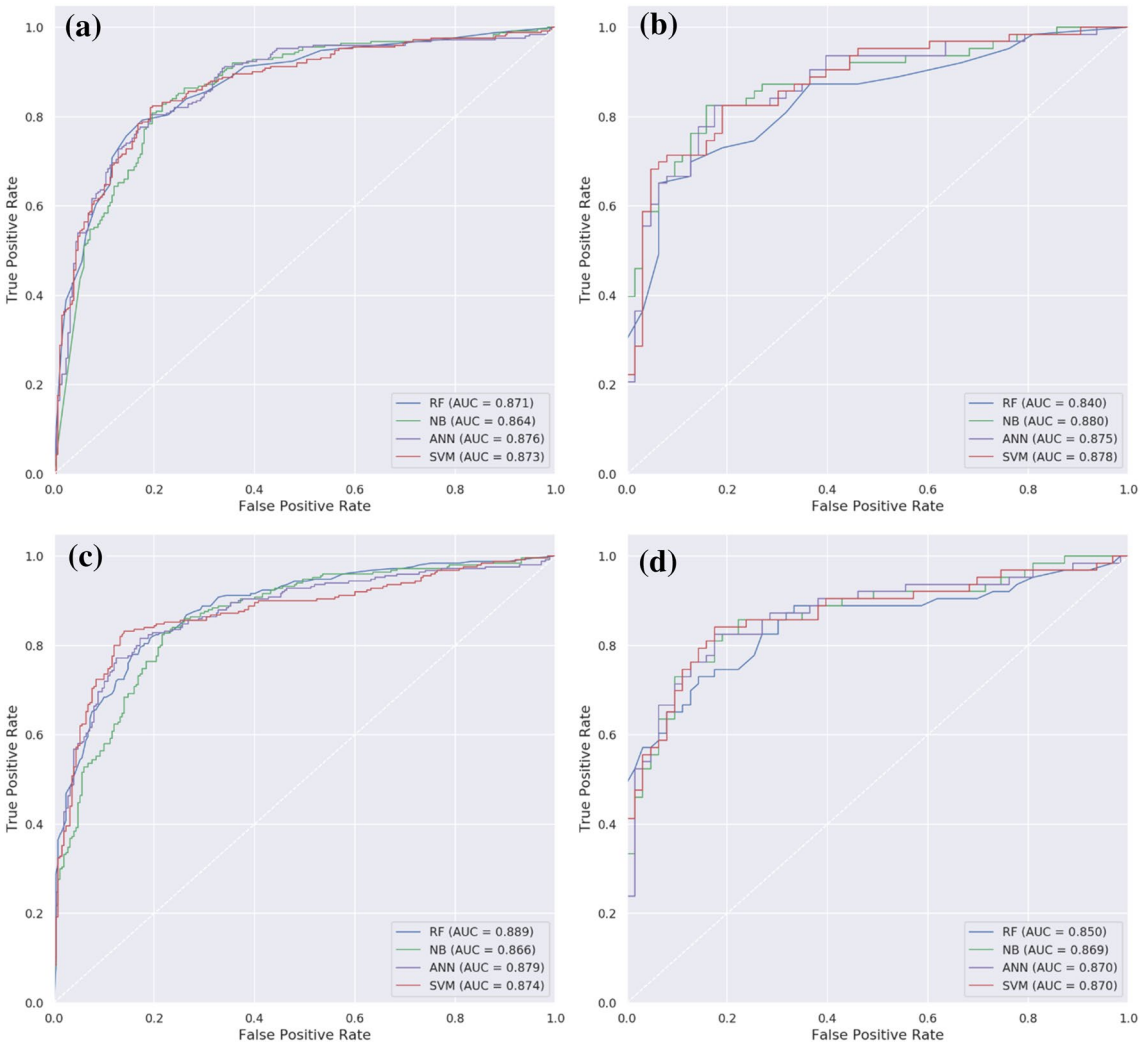


Fig. 4 ROC curves of various meta-predictors developed using RF, NB, ANN and SVM inconjunction with 28D (**a** and **b**) and 16D (**c** and **d**) probabilistic features over tenfold crossvalidation(**a** and **c**) and independent test (**b** and **d**)

Table 5 Comparison of various meta-predictors developed using various classifiers with $m = 16$ informative probabilistic features

Cross-validation	Classifier	Ac	Sn	Sp	MCC	AUC
Tenfold CV	RF	0.812	0.808	0.816	0.632	0.889
	NB	0.792	0.792	0.792	0.593	0.866
	ANN	0.818	0.812	0.824	0.640	0.879
	Meta-iPVP	0.846	0.860	0.832	0.698	0.874
Independent test	RF	0.794	0.857	0.730	0.592	0.850
	NB	0.817	0.873	0.762	0.639	0.869
	ANN	0.802	0.841	0.762	0.605	0.870
	Meta-iPVP	0.817	0.889	0.746	0.642	0.870

Meta-iPVP web server

In order to lower the barrier of access to the predictive model proposed herein, we developed a user-friendly web server called the Meta-iPVP. It is made freely available online at http://camt.pythonanywhere.com/Meta-iPVP. Step-by-step guidelines on using the Meta-iPVP webserver is described hereafter.

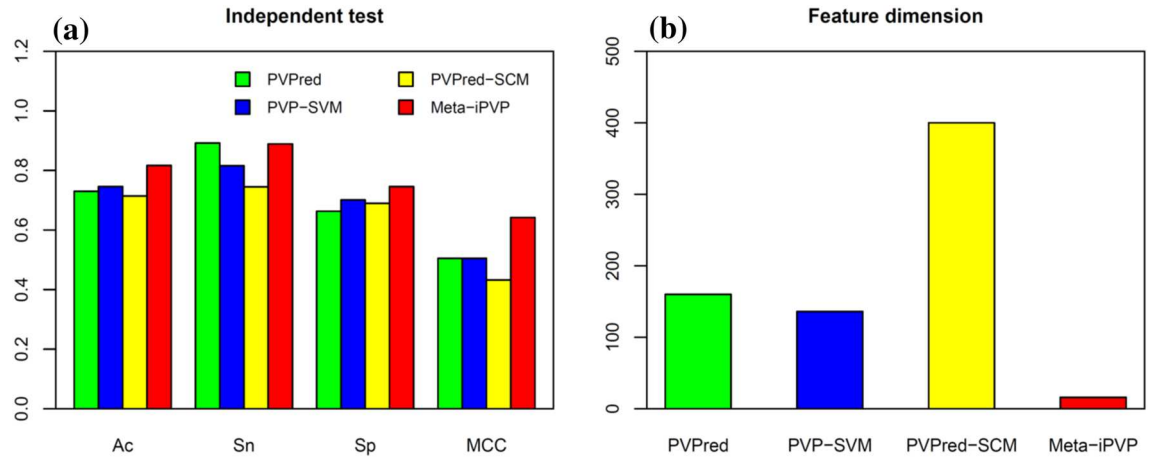


Fig. 5 Comparison of the proposed Meta-iPVP with existing state-of-the-art PVP predictors. **a**Performances of Meta-iPVP and existing PVP predictors in terms of Ac, Sn, Sp and MCC viaindependent test. **b** Number of features used by Meta-iPVP and existing PVP predictors

Table 6 Comparison of independent test results of the proposed predictor with existing state-of-the-art predictors

Methods	Number of feature	Ac	Sn	Sp	MCC	AUC
PVPred[a]	160	0.730	0.892	0.663	0.505	0.857
PVP-SVM[a]	136	0.746	0.816	0.701	0.505	0.844
PVPred-SCM[a]	400	0.714	0.745	0.690	0.432	–
Meta-iPVP	16	0.817	0.889	0.746	0.642	0.870

[a]Results for PVPred, PVP-SVM and PVPred-SCM were obtained by using protein sequences from the PVP-TS dataset as predicted via the web servers of the three aforementioned predictors

(1) Proceed to the Meta-iPVP webserver in an internet browser by going to http://camt.pythonanywhere.com/Meta-iPVP.
(2) Enter the query sequence into the text box or upload the FASTA file by clicking on the "Choose file" button.
(3) Click on the "Submit" button in order to start the prediction process. Typically, the prediction process will take a few seconds.
(4) Results will be shown on the right-hand side of the webserver.

For reference, users can see examples of FASTA-formatted sequences by clicking on the "example file" button.

Conclusions

In this study, we have developed a novel sequence-based meta-predictor called the Meta-iPVP that makes use of probabilistic information for accurate identification of PVPs. In the implementation of the Meta-iPVP, an efficient feature representation learning scheme was used to generate the discriminative probabilistic features based on four machine learning algorithms and seven feature encodings. To obtain an efficient prediction model, the GA-SAR algorithm was used to automatically determine the most informative probabilistic features and for simultaneously tuning the SVM parameters. We further compared the performance of the proposed Meta-iPVP with three well-known and newly developed PVP predictors (PVPred, PVP-SVM and PVPred-SCM). Upon comparing with these state-of-the-art methods, it was found that Meta-iPVP could capture important characteristics for differentiating PVPs from non-PVPs as well as providing improved prediction performances thereby indicating that that the proposed Meta-iPVP is efficient, robust and promising for the identification of PVPs. The predictive model was deployed as a publicly accessible web server so as to allow biologists to make predictions without a technical knowledge of computer science. We believe that the proposed Meta-iPVP webserver can be instrumental in facilitating the identification of novel PVPs for antibacterial drug discovery.

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Compliance with ethical standards

Conflict of interest The authors declare that there are no conflict of interest regarding the publication of this paper.

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